

Topology-Aware Cyberattack Detection in Smart Grids Using Explainable Graph Neural Networks

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Abstract. Smart grids integrate advanced sensing and communication technologies that improve power system efficiency but also increase vulnerability to cyberattacks. Detecting such attacks is challenging because electrical measurements are strongly interconnected through the grid topology. This paper proposes a graph-based cyberattack detection framework using Graph Neural Networks (GNNs). The power grid is modeled as a graph and a dataset is generated from AC power flow simulations of the IEEE-14 bus system using the *pandapower* framework, where multiple attack scenarios are injected into electrical measurements. Three architectures—GAT, GCN, and GraphSAGE—are evaluated. Experimental results show that GraphSAGE achieves the best performance with an F1-score of about 94% and a ROC-AUC close to 0.97. To improve transparency, GNNExplainer is used to identify the buses contributing most to attack detection, providing interpretable insights for smart grid monitoring.

Keywords: Smart Grids · Cyberattack detection · Graph neural networks · Explainable AI · Anomaly detection · Cyber-physical systems.

1 Introduction

Smart grids represent a major evolution of traditional electrical power systems by integrating advanced sensing, communication, and computational technologies into the energy infrastructure. Through the deployment of smart meters, distributed sensors, and automated control mechanisms, smart grids enable real-time monitoring, improved energy efficiency, and enhanced reliability of electricity delivery. However, this increased connectivity and digitalization also significantly expands the attack surface of the power grid. Cyber adversaries may exploit vulnerabilities in communication networks or measurement systems to manipulate grid data, disrupt control operations, or compromise the stability of power system operation. Consequently, the development of reliable cyberattack detection mechanisms has become a critical requirement for secure smart grid management. Cyberattacks in smart grids often target measurement integrity. Malicious actors may manipulate electrical measurements such as voltage, frequency, or power consumption in order to mislead control algorithms or state estimation processes. Among the most dangerous threats are false data injection attacks (FDIAs), measurement manipulation, and coordinated attacks affecting multiple grid components simultaneously. These attacks can remain stealthy while gradually altering the perceived state of the grid, making them difficult to detect using traditional monitoring techniques. Machine learning-based intrusion detection systems have therefore attracted increasing attention for smart grid cybersecurity. Classical approaches such as Support Vector Machines, Random Forest, and Isolation Forest have demonstrated promising capabilities for detecting anomalies

in energy data. However, these models typically treat measurements as independent observations and ignore the structural dependencies between grid components. In reality, electrical networks exhibit strong spatial correlations, where the state of each bus is influenced by neighboring buses through physical power flows. Ignoring these relationships can significantly limit the effectiveness of conventional machine learning methods in detecting coordinated or spatially distributed attacks. Graph-based learning provides a natural framework to model such interconnected systems. In this representation, grid entities such as buses or smart meters are modeled as nodes, while electrical or communication connections form the edges of the graph. Graph Neural Networks (GNNs) have recently emerged as powerful tools capable of learning complex patterns from graph-structured data. By aggregating information from neighboring nodes, GNNs can capture spatial dependencies within the grid and identify abnormal patterns that propagate through the network. Despite their strong modeling capability, deep learning models often suffer from limited interpretability. In safety-critical infrastructures such as smart grids, grid operators require not only accurate detection of cyber incidents but also clear explanations of why a particular event is flagged as suspicious. The lack of interpretability may reduce trust in automated monitoring systems and complicate incident response. Explainable Artificial Intelligence (XAI) techniques aim to address this challenge by providing insights into the decision-making process of complex machine learning models. Motivated by these challenges, this paper proposes an explainable graph-based framework for cyberattack detection in smart grids. The proposed approach models the power grid as a graph and applies Graph Neural Networks to learn topology-aware representations of grid measurements. To evaluate the approach under physically realistic conditions, a graph dataset is generated from AC power flow simulations of the IEEE-14 bus system using the *pandapower* framework. Temporal windows of electrical measurements are used to construct graph samples, while several cyberattack scenarios are injected to emulate realistic disturbances in the grid. Experimental results show that the proposed framework achieves strong detection performance, reaching an F1-score of 94.2% and a ROC-AUC of approximately 0.97 on the IEEE-14 benchmark. Furthermore, explainability analysis using GNNExplainer reveals that the model correctly highlights the buses involved in the injected attacks, providing interpretable insights into the detected anomalies. The main contributions of this work are summarized as follows:

- A realistic graph-based dataset generation pipeline for smart grid cyberattack detection built on AC power flow simulations of the IEEE-14 test system using the *pandapower* framework.
- A temporal graph representation that integrates electrical measurements from multiple operating states, enabling the learning of both spatial and short-term temporal dependencies in the power grid.
- A comparative evaluation of three graph neural architectures (GAT, GCN, and GraphSAGE) for cyberattack detection under physically grounded smart grid conditions.
- An explainability analysis using GNNExplainer that identifies the buses responsible for detected anomalies, providing interpretable insights for smart grid operators.

The remainder of this paper is organized as follows. Section 2 reviews related work on cyberattack detection and graph-based learning in smart grids. Section 3 presents the proposed explainable graph neural network framework. Section 4 describes the experimental setup and evaluation results. Section 5 discusses the findings and implications for smart grid security. Finally, Section 6 concludes the paper and outlines future research directions.

2 Related Work

The increasing integration of communication networks and intelligent monitoring devices in smart grids has significantly improved energy management and operational efficiency. However, this connectivity also exposes the grid to various cyber threats, including false data injection attacks, data manipulation, and coordinated cyber-physical attacks. Consequently, numerous studies have explored machine learning and deep learning techniques for detecting cyberattacks in smart grid environments.

2.1 Machine Learning for Cyberattack Detection in Smart Grids

Machine learning has been widely applied to detect anomalies and cyberattacks in smart grid measurements. Early studies relied on classical supervised learning techniques such as Support Vector Machines (SVM), Random Forest (RF), and decision trees to identify abnormal patterns in electrical measurements such as voltage, frequency, and power consumption. Wang and Lu [10] provided an overview of machine learning approaches for smart grid cybersecurity and highlighted their potential for improving intrusion detection systems.

Several studies have focused on detecting false data injection attacks using machine learning models trained on historical grid measurements. For instance, He and Yan [2] proposed a detection approach based on Random Forest classifiers that analyze variations in power system measurements to identify malicious data manipulation. Similarly, Mukherjee and Roy [5] investigated deep learning techniques for detecting blind cyberattacks in smart grids and demonstrated improved detection accuracy compared to traditional statistical methods.

Unsupervised learning approaches have also been explored to address scenarios where labeled attack data are limited. Ahmed et al. [1] presented a comprehensive survey of anomaly detection techniques in networked systems, highlighting the effectiveness of methods such as Isolation Forest and clustering-based approaches. Furthermore, recent studies have investigated machine learning techniques for detecting electricity theft and abnormal consumption patterns in smart grid environments [7]. Despite their promising results, these approaches generally treat measurements as independent observations and do not explicitly consider the network structure of power systems.

2.2 Graph Neural Networks in Power Systems

Power systems are naturally represented as networks in which buses, substations, and smart meters are interconnected through transmission lines. Graph-based learning methods therefore provide a natural framework for modeling such systems. Graph Neural Networks (GNNs) extend deep learning techniques to graph-structured data by allowing information exchange between neighboring nodes.

The Graph Convolutional Network (GCN) proposed by Kipf and Welling [3] introduced a spectral-based convolution operation for graph data and has become one of the foundational architectures in graph learning. Later, Veličković et al. [9] proposed the Graph Attention Network (GAT), which incorporates attention mechanisms to learn the relative importance of neighboring nodes.

Several surveys have highlighted the growing importance of graph neural networks in modeling complex networked systems [12, 15]. In the context of power systems, GNN-based models have been applied to tasks such as state estimation, fault detection, and operational risk prediction [14]. More recently, Wu and Liu [11] proposed a graph attention-based intrusion detection framework for smart grids, demonstrating the potential of GNN architectures for cybersecurity applications.

Additionally, new datasets and benchmarks have been introduced to facilitate the application of graph learning techniques in power system analysis. For example, Varbella et al. [8] proposed the PowerGraph dataset to evaluate graph neural network models in power grid applications.

2.3 Explainable Artificial Intelligence

Although deep learning models have shown strong performance in anomaly detection tasks, their lack of interpretability remains a major challenge, particularly in safety-critical infrastructures such as smart grids. Explainable Artificial Intelligence (XAI) has therefore emerged as an important research direction to improve transparency and trust in machine learning models.

Several explanation techniques have been proposed to interpret model predictions. Ribeiro et al. [6] introduced the LIME framework, which approximates complex models locally using interpretable surrogate models. Similarly, Lundberg and Lee [4] proposed SHAP values to quantify the contribution of each feature to the model output.

More recently, explanation techniques specifically designed for graph neural networks have been developed. Among them, GNNExplainer [13] identifies the most relevant subgraphs and node features responsible for a particular prediction. Such methods are particularly useful in cybersecurity applications, where understanding the origin of detected anomalies is essential for effective response.

2.4 Research Gap

Despite the significant progress in applying machine learning techniques to smart grid cybersecurity, two main challenges remain. First, many existing approaches fail to exploit the relational structure of grid components and therefore cannot fully capture the spatial dependencies within the network. Second, most deep learning-based detection models operate as black boxes and provide limited interpretability. To address these limitations, this work proposes a graph neural network framework with explainability support for cyberattack detection in smart grids. By integrating graph-based learning with post-hoc explainable AI techniques, the proposed approach captures the structural relationships between grid nodes and provides interpretable indications about the nodes and features that contribute most to attack detection.

3 Methodology

3.1 Power Grid Modeling

To evaluate graph-based cyberattack detection in smart grids, we model the electrical network using the IEEE-14 bus test system. The power grid topology and operating conditions are simulated using the `pandapower` library, which performs AC power flow analysis and provides physically consistent electrical measurements.

In this representation, each bus of the power grid corresponds to a node in the graph, and each transmission line corresponds to an edge connecting two nodes. The resulting graph therefore preserves the physical topology of the electrical network.

For each operating state, a power flow simulation is executed to compute key electrical quantities, including:

- Bus voltage magnitude

- Bus voltage angle
- Active power injection
- Reactive power injection

These measurements form the base features associated with each node in the graph.

3.2 Temporal Graph Construction

Smart grid disturbances and cyberattacks typically evolve over time rather than appearing instantaneously. To capture short-term temporal dynamics, each graph sample is constructed from a sliding window of W consecutive operating states.

Let N denote the number of buses and F the number of electrical features per bus. For a temporal window of size W , the node feature matrix becomes

$$X \in \mathbb{R}^{N \times (F \times W)}$$

where the measurements from each time step are concatenated along the feature dimension.

In the experiments, the IEEE-14 system contains $N = 14$ nodes and the temporal window size is set to $W = 4$, resulting in a total of 28 node features per bus.

Each sample is therefore represented as a graph

$$G = (V, E, X)$$

where V is the set of buses, E the set of transmission lines, and X the node feature matrix.

3.3 Cyberattack Simulation

To evaluate detection capability, several types of cyber-physical disturbances are injected into the simulated measurements. The attack generation process modifies the electrical measurements of randomly selected buses, emulating realistic attack scenarios observed in smart grid infrastructures.

The considered perturbations include:

- **Measurement noise injection:** small random perturbations added to electrical measurements.
- **False data injection:** systematic manipulation of voltage or power values.
- **Load manipulation:** artificial increase or decrease of active and reactive power demand.
- **Measurement inconsistency:** conflicting changes in related electrical variables.

Each generated graph is labeled as either normal or attacked depending on whether such perturbations are present. The final dataset contains 1600 graph samples, which are divided into training, validation, and test sets with proportions of 70%, 15%, and 15%, respectively.

3.4 Graph Neural Network Models

Three graph neural network architectures are evaluated for cyberattack detection:

- **Graph Attention Network (GAT)**, which learns attention coefficients to weight the contribution of neighboring nodes.

- **Graph Convolutional Network (GCN)**, which performs spectral graph convolution based on normalized adjacency matrices.
- **GraphSAGE**, which aggregates neighborhood features using inductive learning mechanisms.

Each model consists of three graph convolution layers followed by a graph-level pooling operation. The node embeddings produced by the convolution layers are aggregated using a combination of mean pooling and max pooling:

$$h_G = [\text{mean}(H) \parallel \text{max}(H)]$$

where H denotes the matrix of node embeddings and $\parallel$ represents concatenation.

The resulting graph representation is then passed to a fully connected layer that outputs the probability of a cyberattack.

3.5 Training Procedure

All models are trained using the Adam optimizer with a binary cross-entropy loss function. Early stopping is applied based on the validation F1-score in order to prevent overfitting. During evaluation, the classification threshold is selected on the validation set to maximize the F1-score. This threshold is then applied to the test set to compute the final performance metrics. The following evaluation metrics are reported: Accuracy; Precision; Recall; F1-score; Area Under the ROC Curve (ROC-AUC). Using a shared dataset and identical training conditions ensures a fair comparison between the evaluated graph neural architectures.

3.6 Explainability with GNNExplainer

To interpret the decision process of the trained models, we apply GNNExplainer, a post-hoc explainability method designed for graph neural networks. GNNExplainer identifies the most influential nodes and edges that contribute to a given prediction. For a target graph G , the method learns a mask over nodes and edges that maximizes the mutual information between the model prediction and the masked graph structure. This allows identification of the subgraph responsible for the classification decision. The node-importance scores produced by GNNExplainer are used to determine which buses have the strongest influence on the attack detection result. This analysis provides insight into whether the model focuses on physically meaningful locations within the grid.

4 Experimental Setup

This section describes the implementation details, model configurations, and training settings used in the experiments. The goal is to ensure reproducibility and to provide a fair comparison between the evaluated graph neural network architectures.

4.1 Dataset Configuration

All experiments are conducted on a simulated dataset generated from the IEEE-14 bus power system using the **pandapower** simulation framework. The dataset contains 1600 graph samples representing different operating states of the grid, including both normal and attacked conditions. Each graph

contains: (i) 14 nodes corresponding to buses; (ii) edges corresponding to transmission lines; (iii) 28 node features obtained from a temporal window of four consecutive operating states. The dataset is divided into three disjoint subsets: Training set: 1120 graphs (70%); Validation set: 240 graphs (15%); Test set: 240 graphs (15%). All models are trained and evaluated using the same dataset split to ensure a fair comparison.

4.2 Model Architecture

Three graph neural network architectures are evaluated: GAT, GCN, and GraphSAGE. Each model follows a similar architectural design consisting of stacked graph convolution layers followed by graph-level pooling. For all models, the architecture includes:

- 3 graph convolution layers
- hidden dimension of 64
- ReLU activation functions
- dropout regularization

The node embeddings produced by the final graph convolution layer are aggregated using both mean pooling and max pooling:

$$h_G = [\text{mean}(H) \parallel \text{max}(H)]$$

where H denotes the matrix of node embeddings. The resulting graph-level representation is passed through a fully connected layer that outputs the probability of a cyberattack. For the GAT model, multi-head attention is used in the first layer to improve neighborhood representation learning.

4.3 Training Configuration

All models are trained using the Adam optimizer with the following hyperparameters: Learning rate: 10^{-3} ; Weight decay: 10^{-4} ; Maximum epochs: 40; Early stopping patience: 8 epochs; batch size: 64. Binary cross-entropy loss is used for training. Early stopping is applied based on the validation F1-score to prevent overfitting. During evaluation, the classification threshold is selected on the validation set to maximize the F1-score. The same threshold is then applied to the test set to compute the final metrics.

4.4 Evaluation Metrics

The performance of the models is assessed using standard classification metrics: Accuracy; Precision; Recall; F1-score; Area Under the Receiver Operating Characteristic Curve (ROC-AUC). These metrics provide complementary perspectives on model performance. In particular, the F1-score is emphasized since it balances precision and recall and is especially relevant in cyberattack detection scenarios where both missed detections and false alarms are critical.

Table 1: Performance comparison on the IEEE-14 cyberattack detection benchmark.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
GAT	0.8958	0.9722	0.8268	0.8936	0.9389
GCN	0.8958	0.8923	0.9134	0.9027	0.9702
GraphSAGE	0.9417	0.9913	0.8976	0.9421	0.9677

4.5 Implementation Details

The models are implemented in Python using the PyTorch and PyTorch Geometric libraries. The power system simulations are generated using **pandapower**. All experiments are executed on a CPU-based environment. Random seeds are fixed during dataset generation and model training to ensure reproducibility. The complete experimental pipeline includes dataset generation, graph construction, model training, evaluation, and explainability analysis using GNNExplainer.

5 Results and Discussion

5.1 Experimental Results

This section evaluates the proposed graph-based cyberattack detection framework on the IEEE-14 benchmark under physically grounded operating conditions generated with **pandapower**. A shared dataset was used for all compared models to ensure a fair comparison. The dataset contains 1600 graph samples, split into 1120 training graphs, 240 validation graphs, and 240 test graphs. Each graph is constructed from a temporal window of four consecutive operating states, and each node is represented by 28 features obtained by stacking electrical measurements across time. Three graph neural architectures were evaluated: Graph Attention Network (GAT), Graph Convolutional Network (GCN), and GraphSAGE. For all models, the classification threshold was selected on the validation set to maximize the F1-score, and the same train/validation/test partition was used across all experiments. Table 1 reports the final test results. Among the evaluated models, GraphSAGE achieved the best overall detection performance, with an accuracy of 94.17%, a precision of 99.13%, a recall of 89.76%, an F1-score of 94.21%, and a ROC-AUC of 0.9677. GCN also achieved strong performance, reaching the highest ROC-AUC of 0.9702, while GAT obtained competitive results but remained slightly inferior in overall discrimination quality.

Fig. 1 presents the ROC curves of the three models. All graph-based architectures clearly outperform random guessing, with AUC values above 0.93. The curves also confirm that GCN and GraphSAGE provide more stable ranking quality than GAT, especially in the low false-positive region, which is important in practical smart-grid monitoring scenarios.

5.2 Discussion

The results show that graph neural modeling is highly effective for cyberattack detection in smart grids when the dataset preserves the physical structure of the network and the temporal evolution of measurements. In particular, GraphSAGE provides the best overall trade-off between decision quality and discrimination capability. Its strong precision indicates that false alarms remain limited, while its high recall confirms that most attack instances are correctly detected. This balance is especially desirable in smart-grid security, where both missed detections and alarm overload may

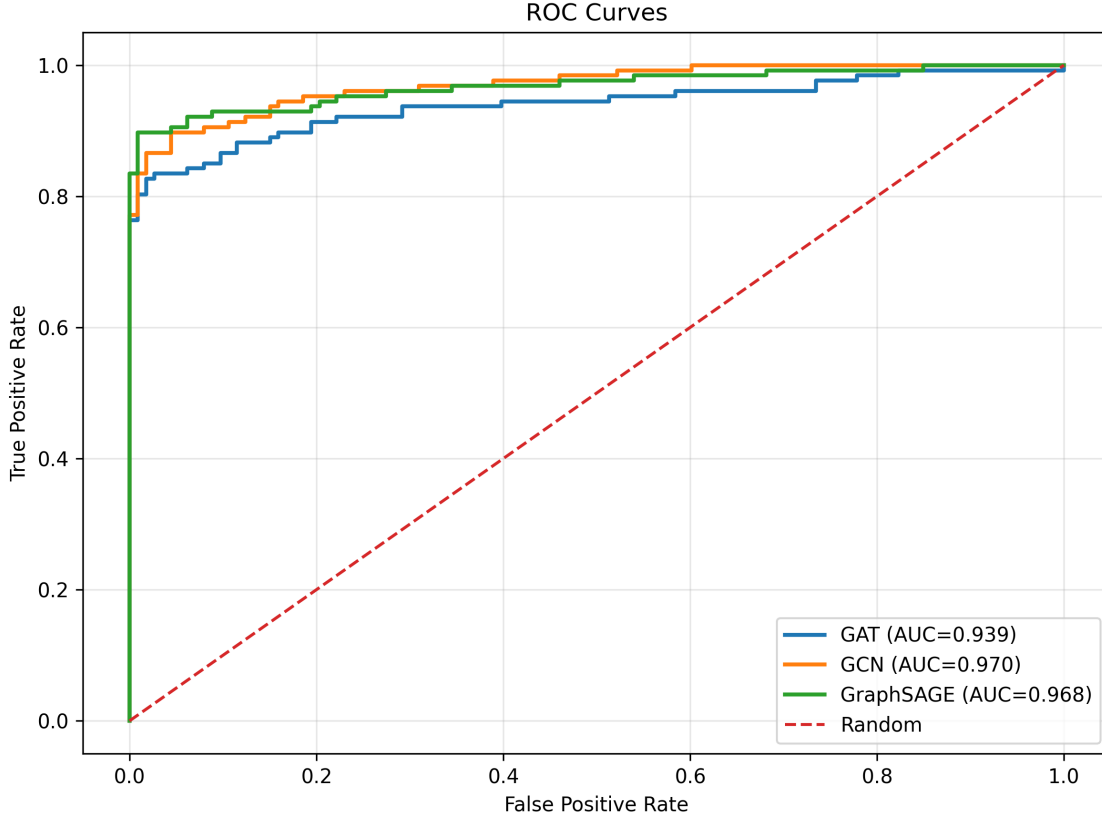


Fig. 1: ROC curves of GAT, GCN, and GraphSAGE on the IEEE-14 cyberattack detection benchmark.

have significant operational consequences. The strong performance of GraphSAGE can be attributed to three main factors. First, its neighborhood aggregation mechanism effectively captures local perturbation propagation across adjacent buses. Second, the use of temporal feature stacking allows the model to exploit short-term attack dynamics rather than relying on a single static snapshot. Third, the combination of mean and max graph pooling helps preserve both global operating trends and localized anomalous patterns, which is particularly relevant for cyber-physical attacks affecting only a subset of buses. GCN also performed very well and even achieved the highest ROC-AUC. This suggests that conventional graph convolution remains a robust choice for ranking normal and attacked states according to confidence. However, GraphSAGE produced better final binary decisions under the optimized threshold, leading to superior accuracy, precision, and F1-score. In contrast, GAT achieved the weakest overall performance among the three architectures. This indicates that attention-based message weighting does not necessarily provide an advantage on a small-scale topology such as IEEE-14, where simpler aggregation mechanisms may already be sufficient to capture the most informative structural dependencies.

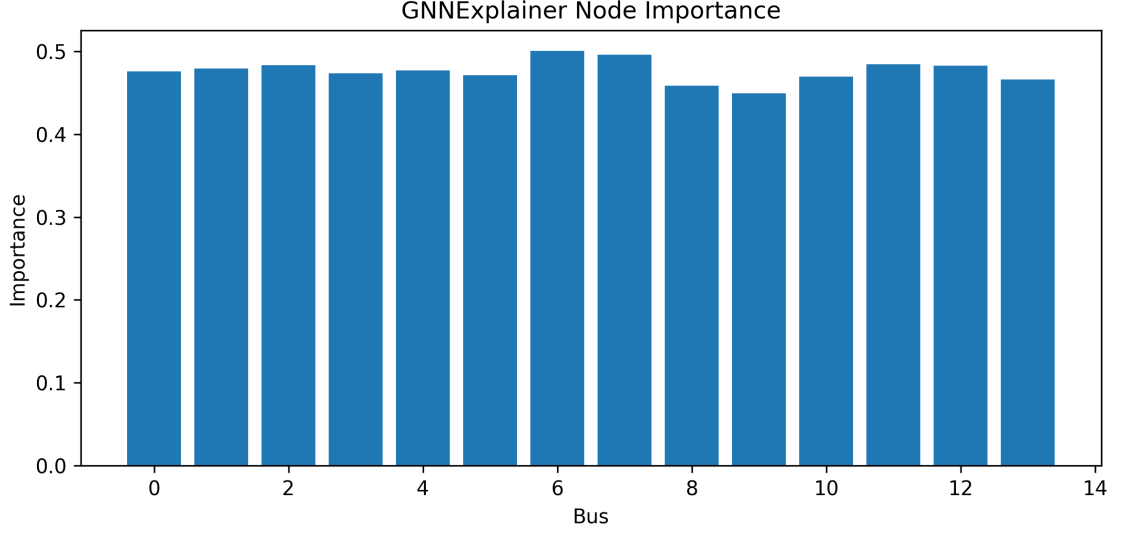


Fig. 2: Node-importance scores produced by GNNExplainer for the best-performing GraphSAGE model.

5.3 Explainability Analysis

To assess whether the best-performing model relies on meaningful electrical patterns, an explainability analysis was conducted using GNNExplainer on the GraphSAGE model. For an *inconsistency* attack under a *stressed* operating regime, the explanation identified buses 6 and 7 as the two most important nodes. These nodes exactly match the ground-truth attacked buses, which provides evidence that the model focuses on attack-relevant regions rather than only on globally dominant buses. Fig. 2 illustrates the node-importance scores produced by GNNExplainer. The highest importance values are assigned to the attacked area, while the remaining buses receive slightly lower but still non-negligible scores, reflecting the fact that disturbances in power grids may propagate beyond the directly attacked nodes. This behavior is consistent with the cyber-physical nature of the problem, where local manipulations can influence neighboring electrical states.

Overall, the explainability results support the reliability of the learned representation. The model not only achieves strong predictive performance but also highlights physically meaningful buses associated with the injected attacks. This improves the practical value of the framework, since interpretable alarms are more useful to grid operators than black-box predictions alone.

5.4 Implications and Limitations

The obtained results demonstrate that graph-based learning is a promising direction for cyberattack detection in smart grids, even on a relatively small benchmark such as IEEE-14. The proposed framework combines physical consistency, temporal measurements, graph topology, and post-hoc explainability, which together form a more realistic evaluation setting than purely synthetic independent feature generation. However, some limitations should be acknowledged. First, the dataset

remains simulation-based, even though it is generated from AC power-flow analysis and realistic operating regimes. Second, the current study is limited to the IEEE-14 topology, which does not fully reflect the scale and complexity of real transmission or distribution systems. Third, only one benchmark setting is considered, and additional experiments on larger networks or real smart-grid datasets would further strengthen the conclusions. Finally, the explainability analysis is demonstrated qualitatively on representative samples; future work could include quantitative localization metrics such as hit rate or overlap between top-ranked explained nodes and attacked buses. Despite these limitations, the results provide strong evidence that GraphSAGE and GCN can accurately detect cyberattacks in physically grounded graph representations of smart grids, while GraphSAGE offers the best overall trade-off between predictive accuracy and interpretability in the current setting.

6 Conclusion

The increasing digitalization of smart grids has significantly improved the efficiency and reliability of modern power systems. However, this transformation has also introduced new cybersecurity challenges, as malicious actors may exploit communication networks or measurement systems to manipulate grid data and disrupt operations. Detecting such cyberattacks is therefore essential for ensuring the security and stability of smart grid infrastructures. In this paper, we proposed an explainable graph-based framework for cyberattack detection in smart grids. The power grid was modeled as a graph structure where buses correspond to nodes and electrical connections represent edges. A dataset was generated from AC power flow simulations of the IEEE-14 bus system using the *pandapower* framework, while several cyberattack scenarios were injected into electrical measurements to emulate realistic disturbances. Temporal windows of measurements were used to construct graph samples that capture short-term system dynamics. Three graph neural architectures—GAT, GCN, and GraphSAGE—were evaluated for attack detection using the same dataset and experimental conditions. The results demonstrate that graph-based learning effectively captures the structural dependencies of the power grid. Among the evaluated models, GraphSAGE achieved the best overall performance, reaching an F1-score of approximately 94% and a ROC-AUC close to 0.97. These results highlight the ability of graph neural networks to detect cyberattacks by exploiting spatial correlations within grid measurements. To enhance the transparency of the proposed detection framework, explainability analysis was conducted using GNNExplainer. The obtained explanations identified the buses that most strongly influenced the detection decision, providing interpretable insights into the detected anomalies. This capability is particularly important for smart grid operators, as it facilitates trust in automated monitoring systems and supports faster incident analysis. Future work will focus on extending the proposed framework to larger power system benchmarks such as IEEE-30 and IEEE-118 networks, integrating real-world measurement datasets, and exploring temporal graph neural networks to capture long-term dynamic behaviors in smart grids. Overall, the proposed approach demonstrates that combining realistic power system simulation, graph neural networks, and explainable artificial intelligence provides a promising direction for reliable and interpretable cyberattack detection in smart grid environments.

References

1. Ahmed, M., Mahmood, A.N., Hu, J.: A survey of network anomaly detection techniques. *Journal of Network and Computer Applications* **60**, 19–31 (2016)

2. He, H., Yan, J.: Detection of false data injection attacks in smart grid using machine learning. *IEEE Access* **9**, 12354–12365 (2021)
3. Kipf, T.N., Welling, M.: Semi-supervised classification with graph convolutional networks. In: *International Conference on Learning Representations (ICLR)* (2017)
4. Lundberg, S.M., Lee, S.I.: A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems* **30** (2017)
5. Mukherjee, D., Roy, S.: Detection of data-driven blind cyber-attacks on smart grids using deep learning. *Sustainable Energy, Grids and Networks* **34**, 100957 (2023)
6. Ribeiro, M.T., Singh, S., Guestrin, C.: Why should i trust you? explaining the predictions of any classifier. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. pp. 1135–1144 (2016)
7. Tsao, Y., et al.: Machine learning techniques for enhancing electricity theft detection in smart grids. *Energy Reports* **10**, 543–556 (2024)
8. Varbella, A., et al.: Powergraph: A benchmark dataset for graph neural networks in power grid applications. *NeurIPS Datasets and Benchmarks Track* (2024)
9. Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., Bengio, Y.: Graph attention networks. In: *International Conference on Learning Representations (ICLR)* (2018)
10. Wang, W., Lu, Z.: Machine learning for cyber security in smart grids: A survey. *IEEE Transactions on Smart Grid* **10**(3), 3462–3473 (2019)
11. Wu, Y., Liu, H.: Graphkan: A graph attention-based intrusion detection framework for smart grids. *Scientific Reports* **15**, 88054 (2025)
12. Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., Yu, P.S.: A comprehensive survey on graph neural networks. *IEEE Transactions on Neural Networks and Learning Systems* **32**(1), 4–24 (2020)
13. Ying, R., Bourgeois, D., You, J., Zitnik, M., Leskovec, J.: Gnnexplainer: Generating explanations for graph neural networks. In: *Advances in Neural Information Processing Systems (NeurIPS)*. vol. 32 (2019)
14. Zhang, Y., Wang, L.: Graph neural networks for power grid operational risk prediction. *Applied Energy* **372**, 123456 (2025)
15. Zhou, J., Cui, G., Zhang, Z., Yang, C., Liu, Z., Wang, L., Li, C., Sun, M.: Graph neural networks: A review of methods and applications. *AI Open* **1**, 57–81 (2020)